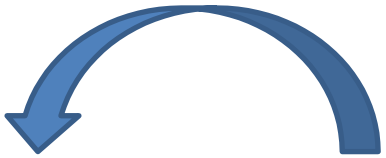
Section 4

Micro CC421

Addressing Modes



Mov destination, source

Table 1-5: Summary of 80x86 Addressing Modes

Addressing Mode	Operand	Default Segment
Register	reg	none
Immediate	data	none
Direct	[offset]	DS
Register indirect	[BX]	DS
	[SI]	DS
	[DI]	DS
Based relative	[BX]+disp	DS
	[BP]+disp	SS
Indexed relative	[DI]+disp	DS
	[SI]+disp	DS
Based indexed relative	[BX][SI]+disp	DS
	[BX][DI]+disp	DS
	[BP][SI]+ disp	SS
	[BP][DI]+ disp	SS

Addressing Modes

1. Register Addressing Mode:

Examples:

```
Mov Ax, Bx
Mov Bx, Dx
Mov Si, Ax
```

Addressing Modes

2. Immediate Addressing Mode:

Examples:

Mov Ax, 122

Mov Bx, 23H

Mov Cl, 1001B

Addressing Modes

3. Direct Addressing Mode:

Examples:

Mov Ax, [12F1]

If DS=2110 then the physical address would be 21100+12F1= 223F1

Square Brackets = Offset of a physical address in memory

1) Calculate the physical address of **Seg.Register:Offset**
2) Read from this address or write into it.

N.B.: in direct addressing mode the DS is the segment register

Addressing Modes

4. Register Indirect Addressing Mode:

Example: Mov [Bx], Ax

Physical address: DS*10H + BX

Mov Cx, [SI]

Physical address: DS*10H + SI

Mov [DI], BX

Physical address: DS*10H + DI

Square Brackets = Offset of a physical address in memory

	Offset	segment register
Register indirect	[BX]	DS
	[SI]	DS
	[DI]	DS

Addressing Modes

5. Based Relative Addressing Mode

Example: Mov Ax, [Bx]+8

Physical address: DS*10H+Bx+8

Mov [bp]+4, CX

Physical address: SS*10H+Bp+4

Square Brackets = Offset of a physical address in memory

	Offset	segment register
Based relative	[BX]+disp	DS
	[BP]+disp	SS

Addressing Modes

6. Indexed Relative Addressing Mode:

Example: Mov Bx, [Di]+2

Physical address: DS*10H+Di+2

Mov [Si]+8, CX

Physical address: Ds*10H+Si+8

Square Brackets = Offset of a physical address in memory

	Offset	segment register
Indexed relative	[DI]+disp	DS
	[SI]+disp	DS

Addressing Modes

7. Based Indexed Relative Addressing Mode:

Example: Mov Ax, [Bx][Si]+8

Physical address: DS*10H+Bx+Si+8

Mov [bp][Di]+4, CX

Physical address: SS*10H+Bp+di+4

	Offset	segment register
Based indexed relative	[BX][SI]+disp	DS
	[BX][DI]+disp	DS
	[BP][SI]+ disp	SS
	[BP][DI]+ disp	SS

Table 1-5: Summary of 80x86 Addressing Modes

Addressing Mode	Operand	Default Segment
Register	reg	none
Immediate	data	none
Direct	[offset]	DS
Register indirect	[BX]	DS
	[SI]	DS
	[DI]	DS
Based relative	[BX]+disp	DS
	[BP]+disp	SS
Indexed relative	[DI]+disp	DS
	[SI]+disp	DS
Based indexed relative	[BX][SI]+disp	DS
	[BX][DI]+disp	DS
	[BP][SI]+ disp	SS
	[BP][DI]+ disp	SS

Assume that the registers have the following values (all in hex) and that CS = 1000, DS = 2000, SS = 3000, SI = 4000, DI = 5000, BX = 6080, BP = 7000, AX = 25FF, CX = 8791, and DX = 1299. Calculate the physical address of the memory where the operand is stored and the contents of the memory locations in each of the following addressing examples.

MOV [SI],AL

Assume that the registers have the following values (all in hex) and that CS = 1000, DS = 2000, SS = 3000, SI = 4000, DI = 5000, BX = 6080, BP = 7000, AX = 25FF, CX = 8791, and DX = 1299. Calculate the physical address of the memory where the operand is stored and the contents of the memory locations in each of the following addressing examples.

MOV [SI],AL

Register indirect, P.A = Ds*10H+Si

P.A= 20000+4000=24000

24000 ← **FF (AL)**

Assume that the registers have the following values (all in hex) and that CS = 1000, DS = 2000, SS = 3000, SI = 4000, DI = 5000, BX = 6080, BP = 7000, AX = 25FF, CX = 8791, and DX = 1299. Calculate the physical address of the memory where the operand is stored and the contents of the memory locations in each of the following addressing examples.

MOV [SI+BX+8],AH

Assume that the registers have the following values (all in hex) and that CS = 1000, DS = 2000, SS = 3000, SI = 4000, DI = 5000, BX = 6080, BP = 7000, AX = 25FF, CX = 8791, and DX = 1299. Calculate the physical address of the memory where the operand is stored and the contents of the memory locations in each of the following addressing examples.

MOV [SI+BX+8],AH

Based Indexed Relative, P.A.= Ds*10H+Bx+Si+8

P.A.= 20000+6080+4000+8

P.A. = 2A088

2A088 ← **25 (AH)**

Assume that the registers have the following values (all in hex) and that CS = 1000, DS = 2000, SS = 3000, SI = 4000, DI = 5000, BX = 6080, BP = 7000, AX = 25FF, CX = 8791, and DX = 1299. Calculate the physical address of the memory where the operand is stored and the contents of the memory locations in each of the following addressing examples.

MOV [BX],AX

Assume that the registers have the following values (all in hex) and that CS = 1000, DS = 2000, SS = 3000, SI = 4000, DI = 5000, BX = 6080, BP = 7000, AX = 25FF, CX = 8791, and DX = 1299. Calculate the physical address of the memory where the operand is stored and the contents of the memory locations in each of the following addressing examples.

MOV [BX],AX

Register indirect, P.A = Ds*10H+BX

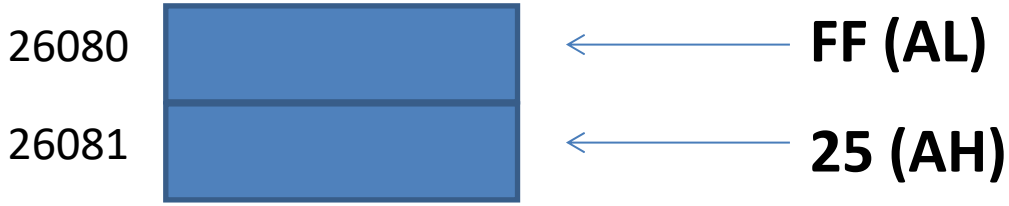
P.A= 20000+6080=26080

Assume that the registers have the following values (all in hex) and that CS = 1000, DS = 2000, SS = 3000, SI = 4000, DI = 5000, BX = 6080, BP = 7000, AX = 25FF, CX = 8791, and DX = 1299. Calculate the physical address of the memory where the operand is stored and the contents of the memory locations in each of the following addressing examples.

MOV [BX],AX

Register indirect, P.A = Ds*10H+BX

P.A= 20000+6080=26080



Assume that the registers have the following values (all in hex) and that CS = 1000, DS = 2000, SS = 3000, SI = 4000, DI = 5000, BX = 6080, BP = 7000, AX = 25FF, CX = 8791, and DX = 1299. Calculate the physical address of the memory where the operand is stored and the contents of the memory locations in each of the following addressing examples.

MOV [DI+6],BX

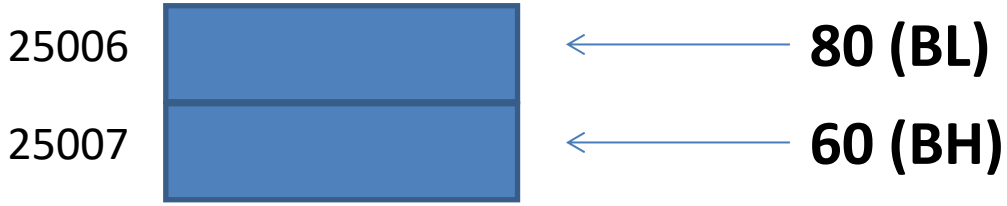
Assume that the registers have the following values (all in hex) and that CS = 1000, DS = 2000, SS = 3000, SI = 4000, DI = 5000, BX = 6080, BP = 7000, AX = 25FF, CX = 8791, and DX = 1299. Calculate the physical address of the memory where the operand is stored and the contents of the memory locations in each of the following addressing examples.

MOV [DI+6],BX

Indexed Relative, P.A.= Ds*10H+Di+6

P.A.= 20000+5000+6

P.A. = 25006



Assume that the registers have the following values (all in hex) and that CS = 1000, DS = 2000, SS = 3000, SI = 4000, DI = 5000, BX = 6080, BP = 7000, AX = 25FF, CX = 8791, and DX = 1299. Calculate the physical address of the memory where the operand is stored and the contents of the memory locations in each of the following addressing examples.

MOV [DI][BX]+28,CX

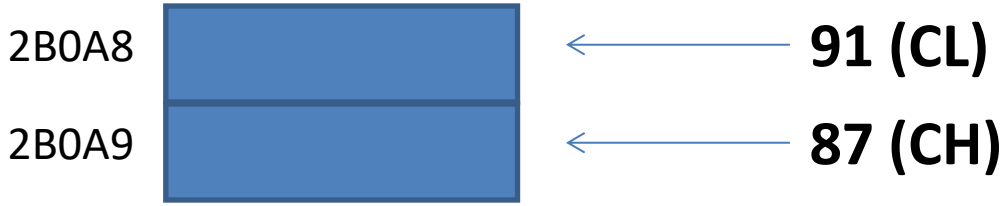
Assume that the registers have the following values (all in hex) and that CS = 1000, DS = 2000, SS = 3000, SI = 4000, DI = 5000, BX = 6080, BP = 7000, AX = 25FF, CX = 8791, and DX = 1299. Calculate the physical address of the memory where the operand is stored and the contents of the memory locations in each of the following addressing examples.

MOV [DI][BX]+28,CX

Based Indexed Relative, P.A.= DSx10H + BX + DI + 28

P.A.= 20000+6080+5000+28

P.A. = 2B0A8



Assume that the registers have the following values (all in hex) and that CS = 1000, DS = 2000, SS = 3000, SI = 4000, DI = 5000, BX = 6080, BP = 7000, AX = 25FF, CX = 8791, and DX = 1299. Calculate the physical address of the memory where the operand is stored and the contents of the memory locations in each of the following addressing examples.

MOV [BP][SI]+10,DX

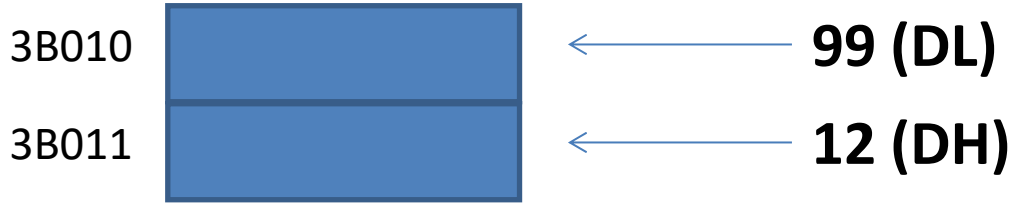
Assume that the registers have the following values (all in hex) and that CS = 1000, DS = 2000, SS = 3000, SI = 4000, DI = 5000, BX = 6080, BP = 7000, AX = 25FF, CX = 8791, and DX = 1299. Calculate the physical address of the memory where the operand is stored and the contents of the memory locations in each of the following addressing examples.

MOV [BP][SI]+10,DX

Based Indexed Relative, P.A.= SS*10+BP+Si+10

P.A.= 30000+7000+4000+10

P.A. = 3B010



Give the addressing mode for each of the following:

(a) MOV AX, DS
(b) MOV BX, 5678H
(c) MOV CX, [3000]
(d) MOV AL, CH
(e) MOV [DI], BX
(f) MOV AL, [BX]
(g) MOV DX, [BP+DI+4]
(h) MOV CX, DS
(i) MOV [BP+6], AL
(j) MOV AH, [BX+SI+50]
(k) MOV BL, [SI]+10
(l) MOV [BP][SI]+12, AX

Addressing mode	Number
Register Addressing Mode	A,d,h
Immediate Addressing Mode	b
Direct Addressing Mode	c
Register Indirect Addressing Mode	E,f
Based Relative Addressing Mode	i
Indexed Relative Addressing Mode	k
Based Indexed Relative Addressing Mode	G,j,l

Show the contents of the memory locations after the execution of each instruction.

(a) MOV BX, 129FH
MOV [1450], BX
DS:1450....
DS:1451 ....

Show the contents of the memory locations after the execution of each instruction.

(a) MOV BX, 129FH
MOV [1450], BX
DS:1450....
DS:1451 ....

DS:1450	9FH
DS:1451	12H